Machine Learning Project Documentation

1. Project Title

Advanced House Price Prediction using Linear Regression

2. Problem Statement

Real estate companies and property buyers often need to estimate the value of a house based on its characteristics such as size, location, quality, and age.

Manually estimating house prices can be inaccurate and time-consuming.
The goal of this project is to build a **Machine Learning model that predicts house prices based on different house features using Linear Regression.**

3. Objective

- Build a regression model to predict house prices.
- Understand how house features influence property prices.
- Perform **data preprocessing and feature engineering**.
- Deploy the model as a **web application using Streamlit**.

4. Dataset Information

The dataset contains various attributes related to houses.

Feature	Description
SalePrice	Target variable representing house price
OverallQual	Overall material and finish quality
GrLivArea	Above ground living area
GarageCars	Number of cars that can fit in the garage
TotalBsmtSF	Total basement area
YearBuilt	Year the house was built
Neighborhood	Location of the house
LotArea	Size of the lot
KitchenQual	Kitchen quality rating

5. Tools Technologies

- Python
- Jupyter Notebook/Vs code
- Pandas
- NumPy
- Scikit-learn
- Matplotlib
- Streamlit
- GitHub

6. Data Preprocessing

Steps performed:

1. Load dataset using pandas
2. Check missing values
3. Encode categorical variables
4. Feature selection
5. Split dataset into training and testing data

Step 1: Load Dataset

The dataset is loaded using Pandas.

- `data = pd.read_csv("train.csv")`

Step 2: Handling Missing Values

Missing values were handled using:

Numeric columns → Median

- `data[num_cols] = data[num_cols].fillna(data[num_cols].median())`

Categorical columns → Mode

- `data[cat_cols] = data[cat_cols].fillna(data[cat_cols].mode().iloc[0])`

Step 3: Encode categorical variables

Categorical variables such as **Neighborhood** and **Kitchen Quality** were converted into numeric values using **One-Hot Encoding**.

- data = pd.get_dummies(data, columns=["Neighborhood","KitchenQual"], drop_first=True)

Step 4: Feature selection

Only important columns were selected for model training.

- OverallQual
- GrLivArea
- GarageCars
- TotalBsmtSF
- YearBuilt
- Neighborhood
- LotArea
- KitchenQual

Step 5: Split dataset into training and testing.

- ✓ X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)

7. Model Building

We used **Linear Regression** to train the model.

Formula:

$$y=mx+cy$$

Where:

- **m** = coefficient (slope)
- **c** = intercept
- **y** = predicted value

Example code:

```
from sklearn.linear_model import LinearRegression

model = LinearRegression() model.fit(X_train, y_train)
```

8. Model Evaluation

Metrics used:

- **RMSE (Root Mean Squared Error)**
  Measures prediction error.
- **R^2 Score**
  Indicates how well the model explains the variance in the data.

Example:

- ✓ rmse = np.sqrt(mean_squared_error(y_test,predicons))
- ✓ r2 = r2_score(y_test, predictions)

9. Model Saving

Model saved using **pickle (.pkl)** format.

```
import pickle

# Save the model
pickle.dump(model, open("house_price_model.pkl", "wb"))
```

10. Deployment

The model was deployed using **Streamlit**.

Example:

```
import streamlit as st
import pickle
# Load saved model
model = pickle.load(open("house_price_model.pkl","rb"))

# User inputs
overallqual = st.slider("Overall Quality",1,10)
grlivarea = st.number_input("Living Area")
garagecars = st.slider("Garage Cars",0,4)

# Prediction
prediction = model.predict([[overallqual,grlivarea,garagecars]])

st.write("Predicted House Price:", prediction)
```

10. Results

The model successfully predicts house prices based on user input features.

Example Prediction:

OverallQual	LivingArea	GarageCars	Prediction
8	2000	2	$210,000

The model provides fast and accurate price estimates.

11. GitHub Repository

Project code available at:

https://github.com/nikhilbontha/House_Price_Prediction_using_ML

12. Streamlit Deployment

Live Application:

https://housepricepredictionusingml-nikhilbontha.streamlit.app/

13. Conclusion

This project demonstrates how **Machine Learning can predict house prices using regression techniques.**

Key achievements:

• Performed **data preprocessing and feature engineering**
• Built a **Linear Regression model**
• Evaluated model performance using RMSE and R^2 score
• Deployed the model as an **interactive Streamlit web application**

Such systems can help **real estate companies, buyers, and sellers estimate property prices efficiently.**

Feature Description:

Feature	Description
SalePrice	Target variable representing the price of the house
OverallQual	Overall quality of the house
GrLivArea	Above ground living area size
GarageCars	Number of cars that can fit in the garage
TotalBsmtSF	Total basement area
YearBuilt	Year the house was built
Neighborhood	Location of the house
LotArea	Size of the lot
KitchenQual	Quality of the kitchen

Example Prediction:

OverallQual	LivingArea	GarageCars	Prediction
8	2000	2	$210,000

The model predicts the house price based on the features entered by the user, providing quick and useful estimates

-- NIKHIL BONTHA